

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2014

Subject: 394171 Mathematics I

Section: 15 - 16

Date: 7 October 2014

Time: 08.00 -10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Textbooks and dictionaries are **NOT** allowed.
4. Calculators are allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 7 pages (including this page), 10 questions and Please do only 9 questions and a total score of 90 marks.
7. Write all your answers on this exam sheet.

Question 1. Given $U = \{1, 2, 3, 4, \dots, 50\}$, $A = \{2, 3, 5, 7, 9, 13\}$, and $B = \{2, 3, 5, 6, 8, 10\}$. Find the set $A - B'$ and the number of element of $P(A - B')$. (10 marks)

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Question 2. Given $U = \{1, 2, 3, 4, 5, \dots, 20\}$, $A = \{2, 3, 5, 7\}$, $B = \{x \mid x = n^2 - 1, n \in I\}$, $C = \{x \in I^+ \mid x > 4\}$, and $D = \{x \mid (x^2 + 6x + 8)x = 0\}$. Find the set $(B \cap A') - (D - C')$. (10 marks)

Question 3. In a family survey asking 1,000 households about their household applications, it was found that each household had at least one of the following applications, refrigerator, iron and stove. It was also found that

385 households have a refrigerator or an iron, but do not have a stove,

535 households have a stove or an iron, but do not have a refrigerator,

400 households have a stove or a refrigerator, but do not have an iron,

255 households have an iron and a stove,

200 households have a refrigerator and a stove,

295 households have a refrigerator and an iron.

How many households have a refrigerator, an iron, and a stove ? (10 marks)

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Question 4. Find the set of all solutions of the inequality $x^2 + 6x + 7 > 2x + 3$.

(10 marks)

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Question 5. Solving the inequality $-1 \leq \frac{2x-1}{2-x} \leq x$. (10 marks)

Question 6 Find an identity of the operator, $a * b = \frac{a + 4b - 3}{4}$, where a, b are real numbers. (10 marks)

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Question 7 Given a, b, c and d are real number and $0 < a < b$ and $0 < c < d$.

To prove that

$$\frac{a}{d} < \frac{b}{c} . \quad (10 \text{ marks})$$

Question 8 Given the relation $r = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x^2 + y^2 \geq 2, y = x^2\}$. Sketch the graph of the relation and find its domain and range. (10 marks)

Question 9 Given $f\left(\frac{2x-1}{2}\right) = 2x+2$, Evaluate the value of $f^{-1}(-2)$ and find $f^{-1}(2x)$. (10 marks)

Question 10 Given $f(2x-1) = 4x+3$ and $g\left(\frac{x}{2}\right) = 4-x$. Find $(f-g)(x)$ and $(f \cdot g)(x-1)$. (10 marks)

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